

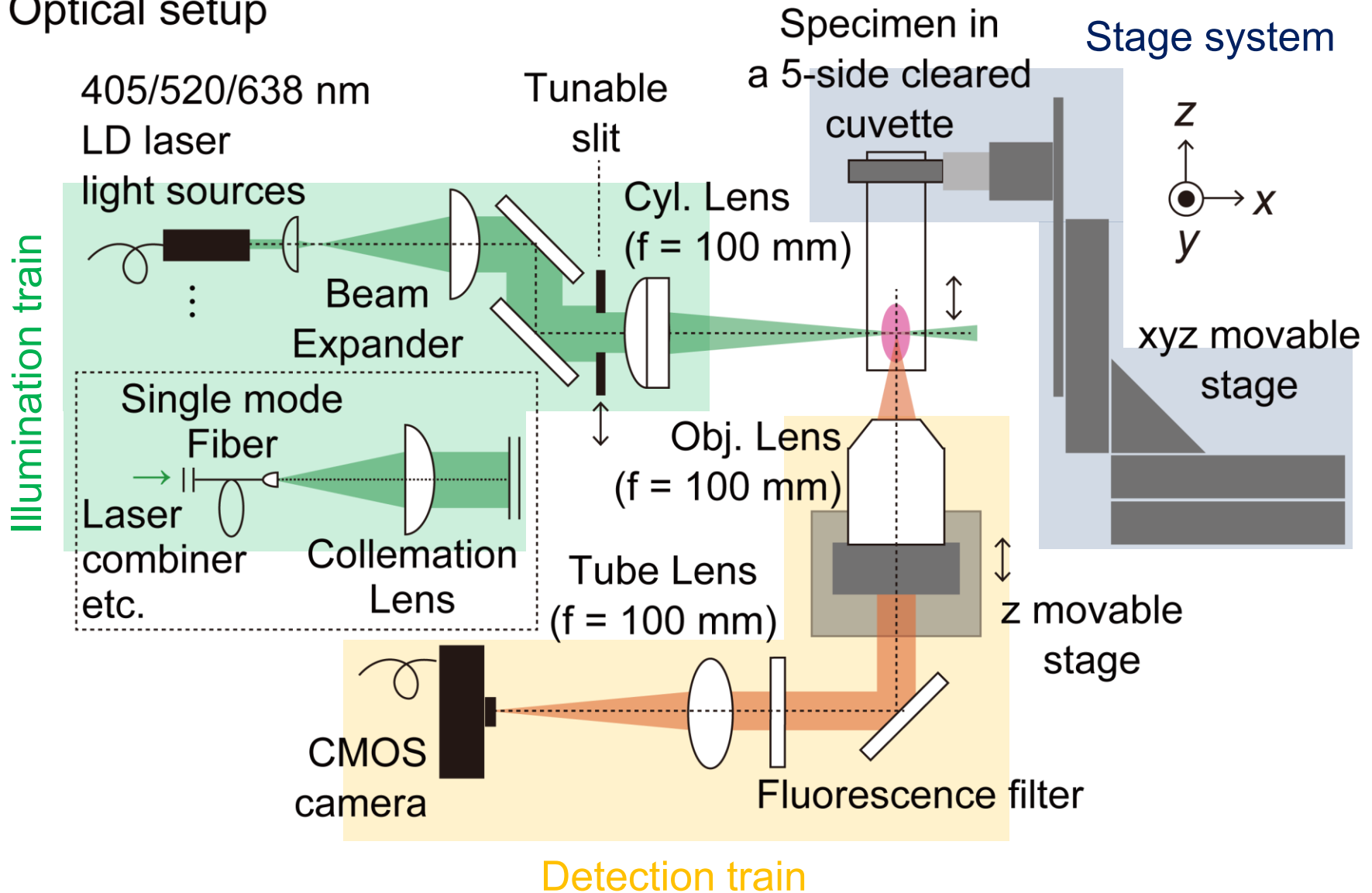
Construction manual of descSPIM-SLIM

(ver. 260414)



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- Optical setup



descSPIM-SLIM parts list

Parts list of descSPIM slim (ver. 20260218)

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Total	¥2,341,912
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Catalog ID	Product	Quantity	Price for one	Subtotal
Base				
MB2025/M	250 x 200 mm breadboard	1	¥22,854	¥22,854
PH100/M	100 mm post holder	2	¥1,565	¥3,130
TR100/M	100 mm post	2	¥978	¥1,956
RLA075/M	75 mm rail	1	¥4,962	¥4,962
Illumination train				
PL255	Compact laser module, 405 nm	1	¥33,636	¥33,636
PL251	Compact laser module, 520 nm	1	¥31,325	¥31,325
PL252	Compact laser module, 635 nm	1	¥24,986	¥24,986
KAD11F	Laser module adaptor	1	¥11,230	¥11,230
CP33/M	SM1-30 mm cage plate	1	¥2,799	¥2,799
C061TMD-A	Aspheric lens, f = 11 mm	1	¥12,982	¥12,982
CP1M09/M	M09 x 0.5-30 mm cage plate	1	¥6,286	¥6,286
AC254-100-A-ML	Achromatic lens, f = 100 mm	1	¥17,066	¥17,066
CP33C/M	30 mm cage bracket	2	¥3,588	¥7,176
ER8-P4	203.2 mm rods (4x)	1	¥7,049	¥7,049
KCB1EC/M	Right-angle mirror mount for 30 mm	2	¥32,665	¥65,330
BBE1-E02	Elliptical mirror, 400-750 nm	2	¥19,116	¥38,232
ER05-P4	12.7 mm rods (4x)	1	¥3,043	¥3,043
VA100CP/M	Adjustable slit for 30 mm cage	1	¥46,321	¥46,321
CP35/M	25 mm lens-30 cage plate	1	¥3,156	¥3,156
ACY254-100-A	Cylindrical achromatic lens, f = 100 mm	1	¥65,476	¥65,476
ER4-P4	101.6 mm rods (4x)	1	¥4,277	¥4,277
Stage system				
XR25P/M	25 mm axial stage	3	¥73,687	¥221,061
XR25-XZ/M	Stage assembly kit	1	¥24,316	¥24,316
ZST225B	25 mm stepping motor actuator	1	¥170,307	¥170,307
KST201	Actuator controller	1	¥113,265	¥113,265
KPS201	Power supply	1	¥5,912	¥5,912
RLA300/M	300 mm dovetail rail	1	¥13,019	¥13,019
RC1	Rail carrier	1	¥4,335	¥4,335
PH20/M	20 mm post holder	1	¥1,197	¥1,197
TR20/M	20 mm post	1	¥789	¥789
KM05S/M	12.7 mm square mount	1	¥11,712	¥11,712

Detection train				
TL2X-SAP	2x objective lens	1	¥208,024	¥208,024
SM1A12	Adapter with External SM1, Internal M25	1	¥3,518	¥3,518
CT1B/M	13 mm z-axial stage	1	¥74,358	¥74,358
ZFS25B	25 mm small stepping motor actuator	1	¥201,302	¥201,302
KST201	Actuator controller	1	¥113,265	¥113,265
KPS201	Power supply	1	¥5,912	¥5,912
ER4-P4	101.6 mm rods (4x)	1	¥4,277	¥4,277
KCB1EC/M	Right-angle mirror mount for 30 mm	1	¥32,665	¥32,665
BBE1-E02	Elliptical mirror, 400-750 nm	1	¥19,116	¥19,116
ER05-P4	12.7 mm rods (4x)	1	¥3,043	¥3,043
RCA7	Rail-30 mm cage plate	1	¥14,089	¥14,089
SM1QP	Filter holder tube	1	¥27,688	¥27,688
FBH450-40	Bandpass Filter CWL = 450 nm, FWHM = 40 nm	1	¥26,553	¥26,553
FBH550-40	Bandpass Filter CWL = 550 nm, FWHM = 40 nm	1	¥26,553	¥26,553
FBH670-40	Bandpass Filter CWL = 670 nm, FWHM = 40 nm	1	¥26,553	¥26,553
SM1A2	SM1-SM2 adaptor	2	¥4,163	¥8,326
TTL100-A	Tube lens, f = 100 mm	1	¥135,804	¥135,804
SM1L10	SM1 tube, 25.4 mm	1	¥2,259	¥2,259
SM1V05	Adjustable SM1 tube	1	¥4,796	¥4,796
SM1P2	SM1 adaptor mount	1	¥4,344	¥4,344
SM1AB2	SM1 adaptor	1	¥3,046	¥3,046
SM1L03	SM1 tube, 25.4 mm	1	¥1,928	¥1,928
SM1M05	SM1 tube, 25.4 mm	1	¥2,252	¥2,252
LP126MU/M	Monochrome CMOS	1	¥387,877	¥387,877
Others				
TC3/M	Metric Hex wrenches	1	¥14,682	¥14,682
BD-3/32	3/32" Balldriver	1	¥662	¥662
VRC2CPT	30 mm Cage System Alignment Plate	1	¥5,269	¥5,269
SPW602	Spanner Wrench for SM1-Thread	1	¥4,566	¥4,566
	M4 and M6 screws			

*depends on fluorophore

*depends on fluorophore

*depends on fluorophore

*optional

*optional

*optional

*optional

*If you don't have

*If you don't have

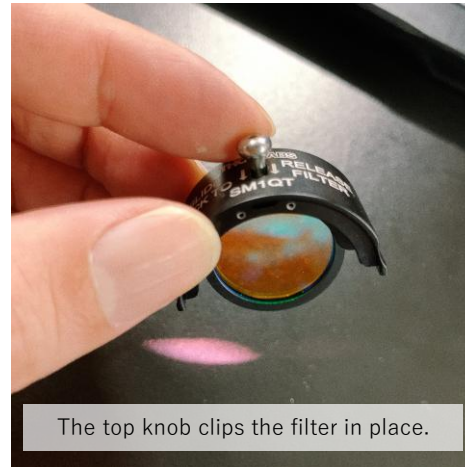
*optional

Prep. before constructions

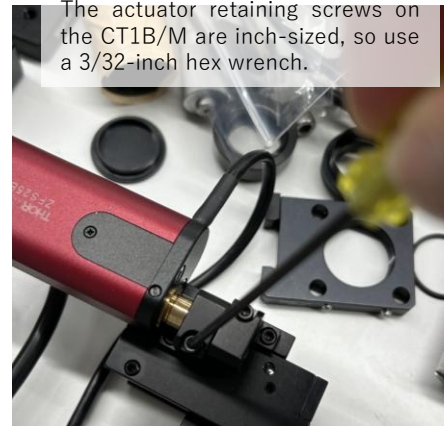
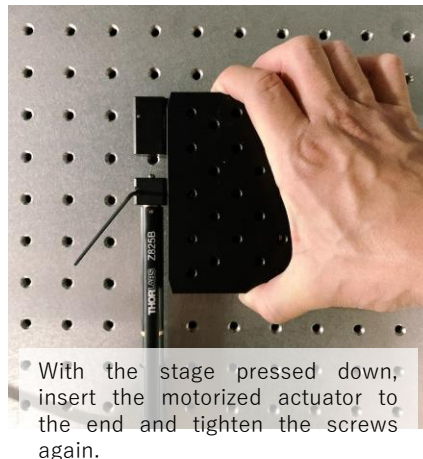
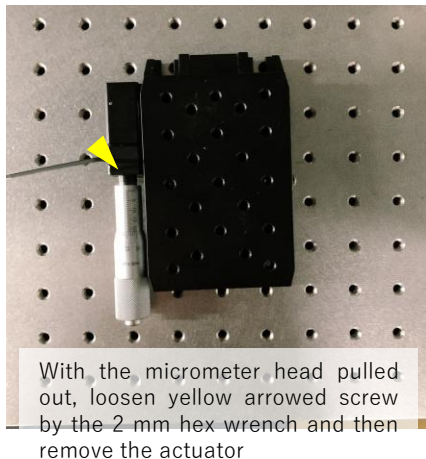
- Connect CP33M and KM05S on $\phi 12$ mm post (photo is other's)



- Set emission filter to filter exchanger carriage (SM1QT)



- Replace the actuators of single-axis stages (XR25P/M, CT1B/M) from micrometer heads to motorized ones (ZST225B, ZFS25B)



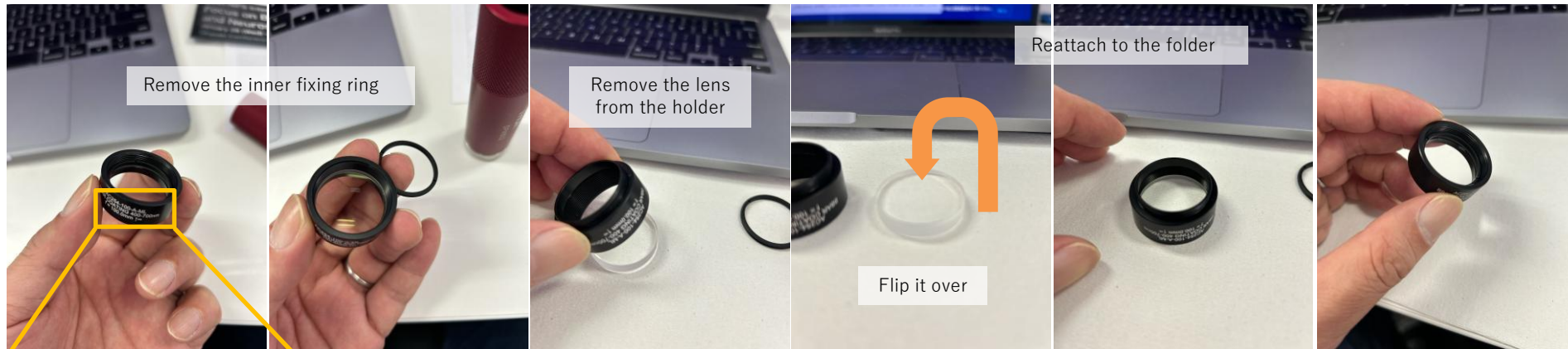
* Caution!

The Travel range of CT1B/M is 13 mm, whereas the range of ZFS25B is 25 mm. Forcing the actuator beyond its travel range may cause operational errors or damage. Confirm the home position and operate within a range not exceeding 13 mm.

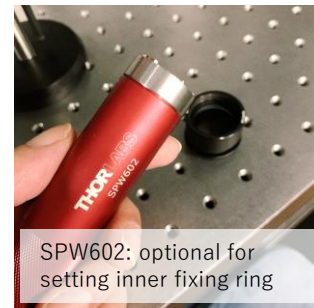
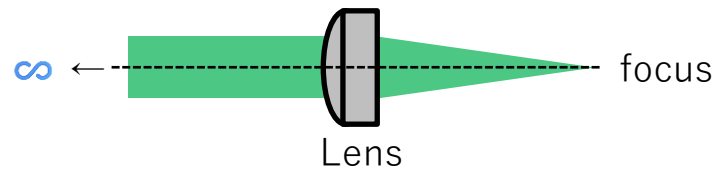


Prep. before constructions

- Direction of Achromatic lens (AC254-100-A-ML) should be inverted against its holder
 * non-inverted direction with additional SM1 cage plate (CP33/M) also OK



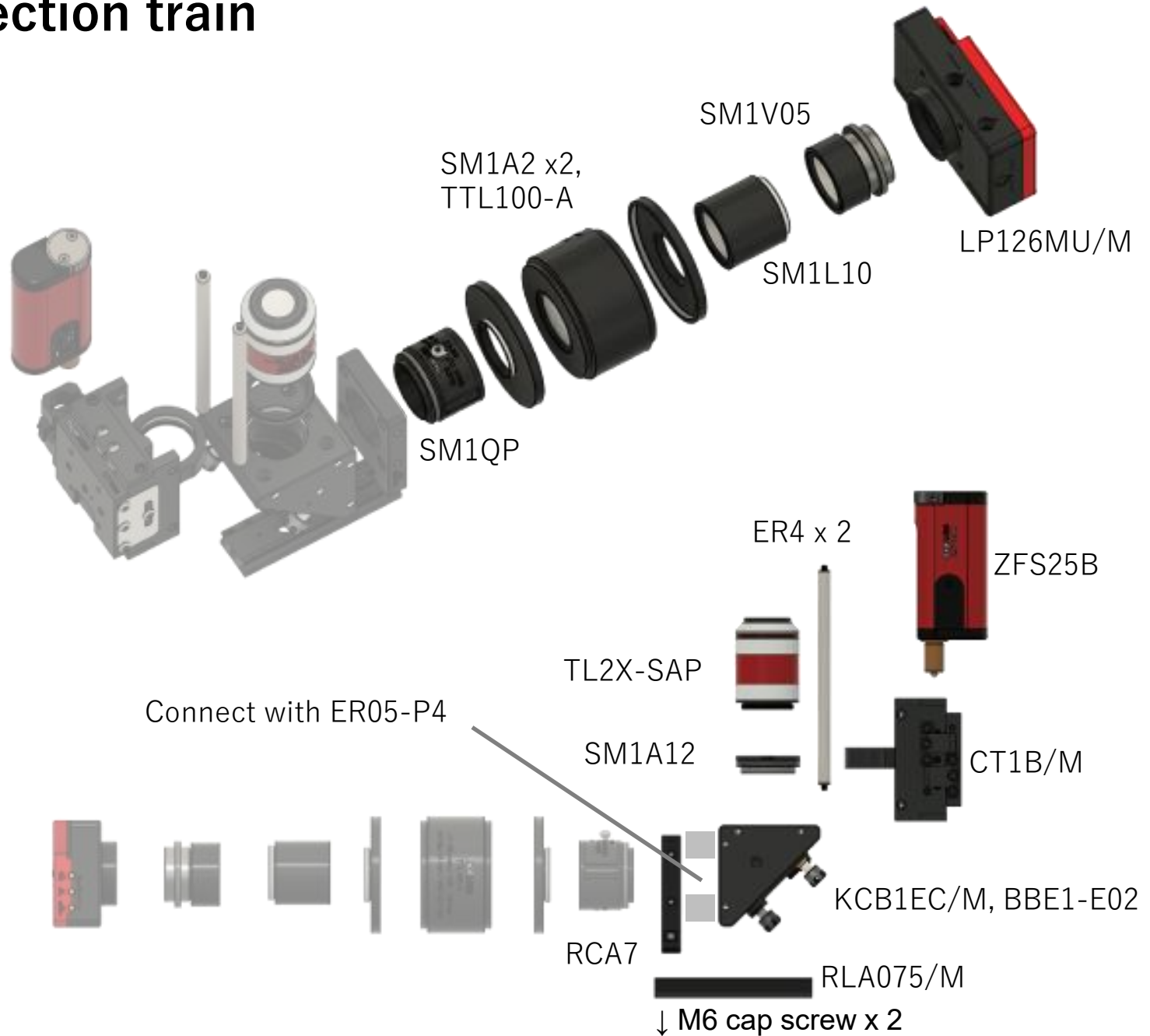
“↑ ∞” means
the direction opposite to focus



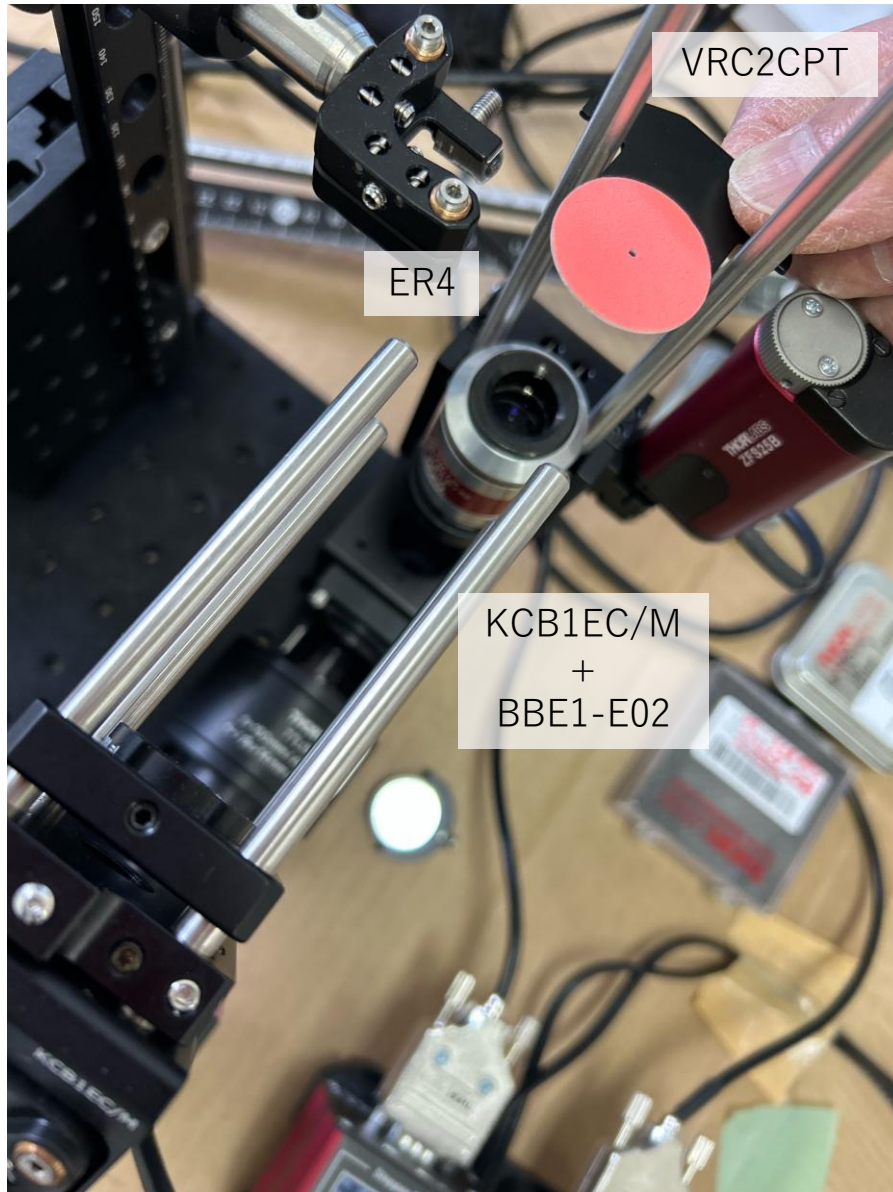
- Set elliptical mirrors (BBE1-E02) to
mirror mounts (KCB1EC/M) (x 3)



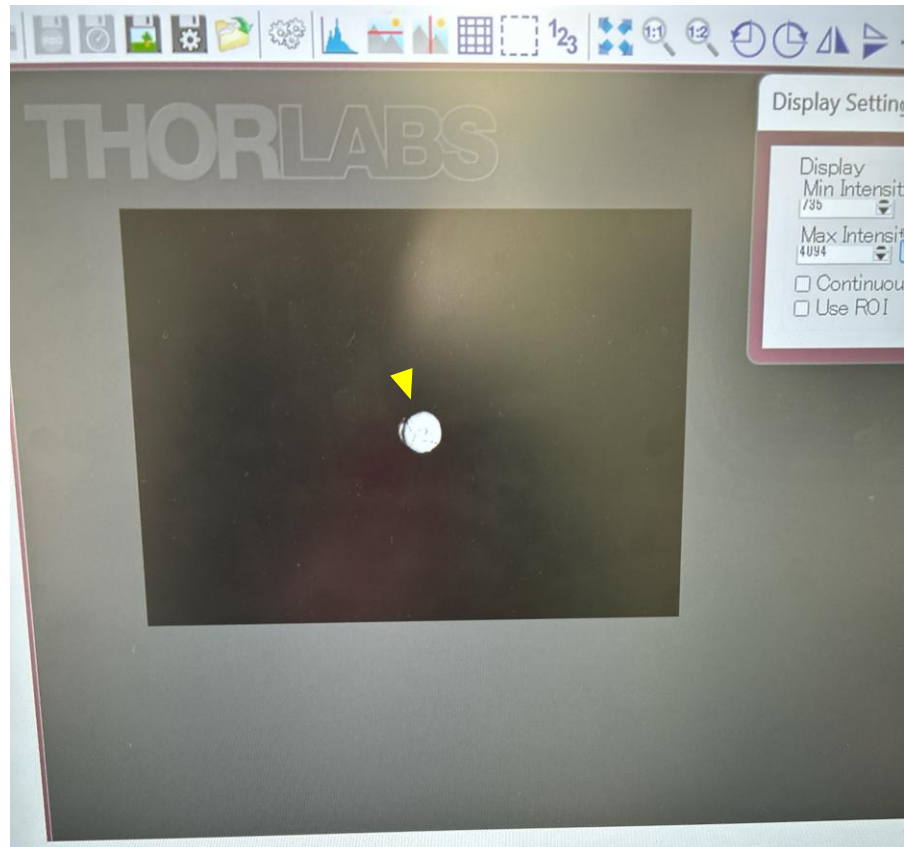
Detection train



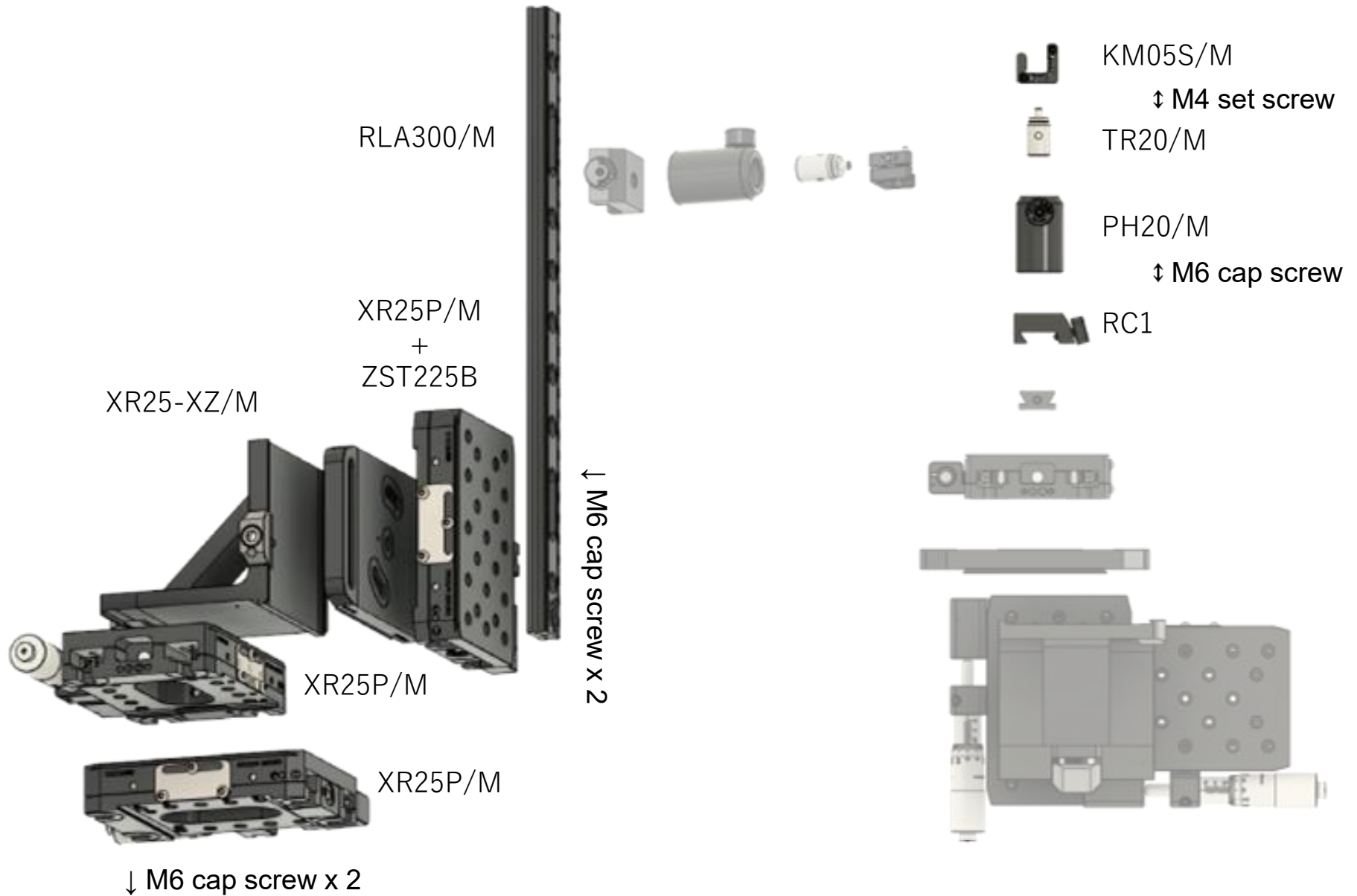
Optical alignment of the detection train



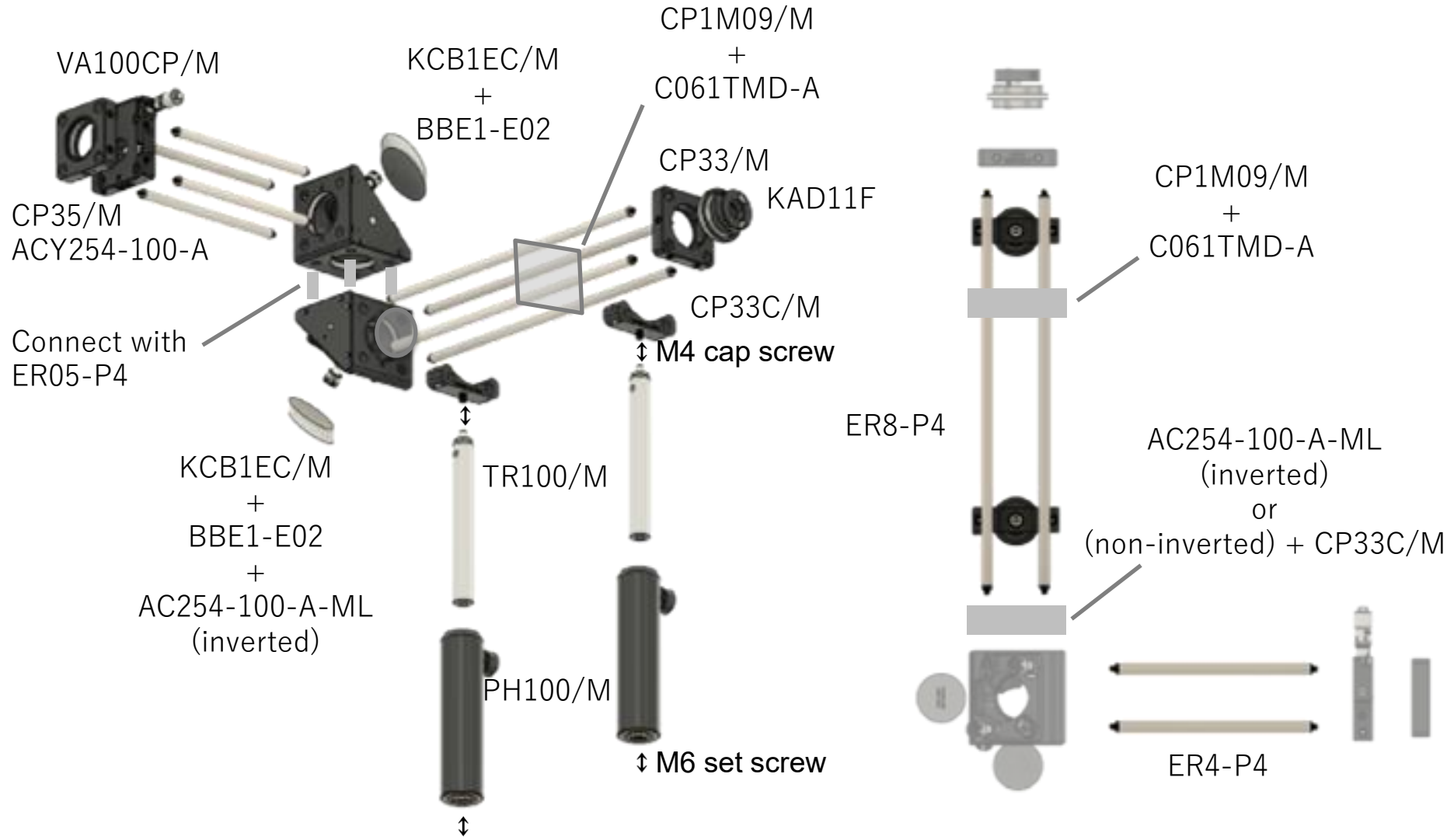
- ↓ Connect the surplus ER4 x 2 to the ER4 x 2 to extend it
- ↓ Start the camera while positioning an alignment tool VRC2CPT (different alignment tool in photo) near the objective lens focal point
- ↓ Adjust the screws on the KCB1EC/M mirror holder so the central aperture is centered in the field of view



Stage system

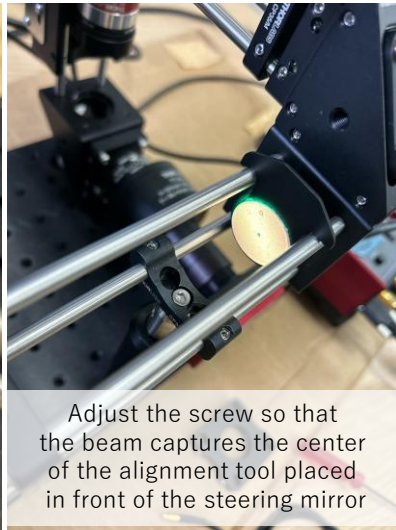


Illumination train

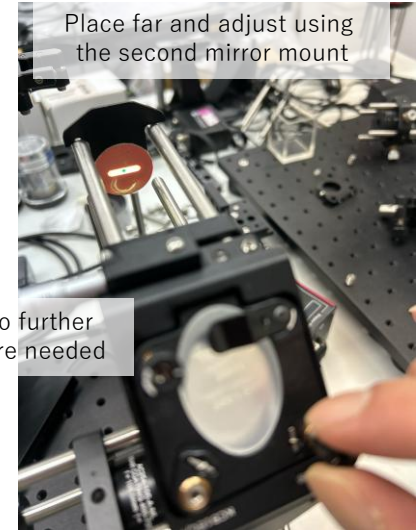


Optical alignment of the illumination train

- Laser module adaptor CD35/M alignment

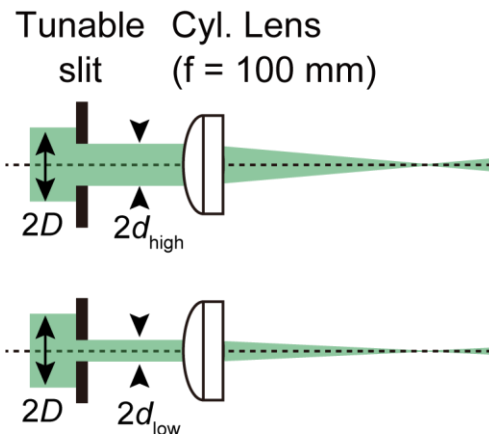
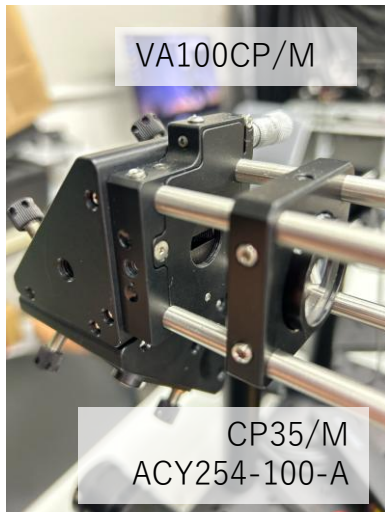


- Steering mirrors



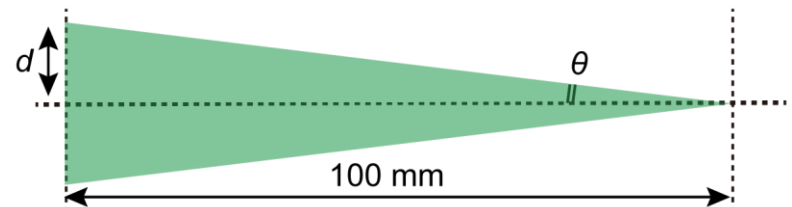
Repeat until no further adjustments are needed

Adjustable NA_{ex} by a horizontal slit



$2D$: Entrance beam diameter (corresponds to $eFOV_y$)

$2d$: Slit height



$NA_{ex} = n \times \sin\theta \doteq d/100$ $\rightarrow$ tunable beam waist radius, Rayleigh length ($eFOV_x$), Axial resolution

Assembling on the breadboard

Connect to PH100/M with M6 set screws (Illumination train)

Connect to XR25P/M with M6 cap screws (Sample holder)



Connect to RLA075/M
with M6 cap screws
(Detection train)